

PETROVSKIY ZAVOD, Russian Federation

WMO: 308380

Lat: 51.317N

Lon: 108.867E

Elev: 2644

StdP: 13.35

Time Zone: 9.00 (E09)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-37.9	-33.7	-41.1	0.6	-37.6	-37.2	0.7	-33.5	14.5	2.6	13.1	1.4	0.7	10	0.670

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	25.7	84.9	62.1	81.5	61.2	78.1	59.9	65.4	78.2	63.8	75.6	62.4	73.5	6.2	160

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
61.4	89.7	68.7	59.9	84.9	67.7	58.3	80.1	67.0	31.9	78.5	30.6	75.7	29.5	73.5	71.4

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4)	14.0	12.2	10.8	DB	-43.6	92.1	5.5	3.7	-47.5	94.8	-50.7	96.9	-53.8	99.0	-57.8	101.7
(5)				WB	-43.3	68.3	5.3	1.8	-47.1	69.6	-50.1	70.6	-53.1	71.6	-56.9	72.9

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	28.1	-11.6	-3.6	14.9	33.4	45.0	58.2	63.4	58.5	45.7	29.2	8.5	-6.4
	DBStd	27.39	9.64	10.25	11.96	8.32	7.18	7.08	5.49	6.15	7.23	8.86	11.44	11.10
	HDD50	9016	1909	1502	1088	498	189	12	0	10	167	645	1246	1750
	HDD65	13542	2374	1922	1553	947	620	220	93	212	580	1110	1696	2215
	CDD50	1023	0	0	0	1	35	260	417	272	38	0	0	0
	CDD65	73	0	0	0	0	1	17	45	10	0	0	0	0
	CDH74	1747	0	0	0	1	64	545	838	264	35	0	0	0
	CDH80	466	0	0	0	0	13	142	261	47	3	0	0	0
Wind	WSAvg	4.3	3.1	3.2	4.5	5.4	5.8	5.1	4.4	4.3	4.4	3.9	3.6	3.2
Precipitation	PrecAvg	12.9	0.2	0.2	0.2	0.4	1.0	1.9	3.0	3.3	1.8	0.5	0.4	0.3
	PrecMax	19.8	0.4	0.3	0.8	1.4	3.7	4.3	5.6	6.2	4.1	1.2	0.6	0.5
	PrecMin	10.3	0.0	0.0	0.0	0.0	0.2	0.3	1.3	1.1	0.6	0.0	0.1	0.0
	PrecStd	2.2	0.1	0.1	0.2	0.4	0.7	0.9	1.3	1.3	0.9	0.3	0.1	0.1
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	18.4	32.1	51.0	69.2	81.7	89.8	91.3	85.2	78.7	63.2	40.4	23.4
		MCWB	16.4	26.1	37.5	47.0	55.0	61.8	66.4	63.3	57.6	46.2	33.1	20.7
	2%	DB	11.5	24.0	44.2	61.6	74.7	83.7	86.1	80.4	72.0	55.4	34.5	18.5
		MCWB	9.7	20.3	33.8	43.8	52.1	60.2	63.9	61.7	54.8	42.3	29.3	16.8
	5%	DB	7.6	19.2	39.3	55.9	69.5	80.3	82.5	76.7	66.5	49.6	30.4	14.5
		MCWB	6.1	16.2	30.8	40.6	49.3	59.5	62.8	61.2	52.3	39.2	26.6	13.0
	10%	DB	3.9	14.9	34.2	50.7	64.0	76.4	78.6	72.7	61.5	44.1	26.5	11.1
		MCWB	2.8	12.5	27.8	38.1	46.5	58.3	61.8	59.6	50.2	36.4	23.7	10.0
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	16.1	26.4	37.9	47.8	56.0	65.4	68.3	66.3	59.6	47.1	33.4	21.0
		MCDB	18.1	31.5	49.3	65.8	78.7	80.7	82.9	76.9	73.9	62.7	39.4	23.0
	2%	WB	9.8	20.6	34.2	44.3	53.1	63.4	66.3	64.0	56.5	43.1	29.8	16.9
		MCDB	11.4	24.0	43.5	60.0	72.1	77.2	80.2	74.2	68.3	54.4	34.2	18.4
	5%	WB	6.2	16.3	31.2	41.1	50.3	61.6	64.8	62.7	53.9	39.8	26.8	13.1
		MCDB	7.5	19.2	38.6	54.9	66.9	74.9	77.3	72.3	63.9	48.2	30.2	14.4
	10%	WB	2.8	12.6	28.2	38.4	47.6	59.7	63.4	61.2	51.2	36.8	23.8	10.0
		MCDB	3.8	15.0	33.9	49.5	61.7	72.5	75.0	70.4	60.6	43.7	26.6	11.1
Mean Daily Temperature Range	5% DB	MDBR	23.4	31.0	28.9	25.0	28.2	28.1	25.7	24.1	25.3	24.2	23.3	20.8
		MCDBR	23.1	33.7	32.6	37.2	41.5	37.3	34.9	33.6	37.7	34.2	24.8	18.8
		MCWBR	21.5	29.6	22.0	20.7	21.0	16.2	14.6	16.2	20.6	21.5	20.0	17.3
	5% WB	MCDBR	22.9	33.0	30.5	34.1	36.8	28.7	27.2	25.9	32.2	31.1	23.8	18.7
		MCWBR	21.5	29.1	20.6	19.6	18.9	13.3	12.5	13.3	18.1	19.8	19.4	17.3
Clear-Sky Solar Irradiance	taub	0.263	0.286	0.319	0.373	0.450	0.438	0.411	0.377	0.331	0.303	0.285	0.267	
	taud	2.177	2.162	2.205	2.216	2.013	2.165	2.298	2.376	2.454	2.450	2.289	2.159	
	Ebn at Noon	237	264	277	273	255	260	266	270	273	259	228	212	
	Edh at Noon	23	31	36	40	52	45	39	34	28	23	20	20	
All-Sky Solar Radiation	RadAvg	443	778	1222	1568	1815	1918	1781	1507	1164	759	444	326	
	RadStd	23	34	47	113	118	113	124	70	100	57	35	23	

Historical Trends

		DBAvg	Heating			Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP		HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	
(47)	Regional (9 neighbors)	N/A	N/A	-2.59	N/A	N/A	N/A	N/A	N/A	N/A	N/A	

Nomenclature: See separate page